

Quiz 5

Tick the statements that are true and leave unticked those that are false.
Each statement is worth 1 point for a correct answer.

Your name: _____

- ☒ $\lambda x. \lambda y. x \equiv_{\alpha} \lambda y. \lambda x. y$
- ☐ $\lambda x. \lambda y. y \equiv_{\alpha} \lambda y. \lambda x. y$
- ☒ $(\lambda x. x) (\lambda y. y) z \longrightarrow_{\beta}^* z$
- ☐ $(\lambda x. y) z \longrightarrow_{\beta}^* z$
- ☒ For every lambda term A , we have $A \longrightarrow_{\beta}^* A$
- ☐ ‘Lambda’ stands for “Logical Apparatus for Modeling Binding and Dynamic Application”.
- ☐ Every lambda term has a normal form.
- ☒ If a term has a normal form, then it’s unique, up to renaming of bound variables.
- ☒ Let $\bar{3}$ be the Church numeral for 3. Then $\bar{3} F X \longrightarrow_{\beta}^* F(F(FX))$.
- ☒ Let $F = \lambda x. xx$ and $G = \lambda x. \lambda y. yx$. Then $GzF \longrightarrow_{\beta}^* zz$.

Bonus 1. Let TRUE and FALSE be the Church Booleans. Describe the term TRUE FALSE in English. (Is it a variable? Is it a function? If so, what does it take and what does it return?)

Answer: A function that takes any argument and returns FALSE.

Bonus 2. Construct terms A , B , C , such that $(AB)C$ and $A(BC)$ have different normal forms.

Answer: For example, $A = \text{TRUE}$ and $B = C = \text{ID}$.

What’s your favourite Greek letter?

Explanations

- ⊗ $\lambda x.\lambda y.x \equiv_\alpha \lambda y.\lambda x.y$. Renaming the outer parameter to u and the inner one to v turns both terms into $\lambda u.\lambda v.u$.
- $\lambda x.\lambda y.y \equiv_\alpha \lambda y.\lambda x.y$. The left term returns its second argument; the right term returns its first.
- ⊗ $(\lambda x.x)(\lambda y.y)z \longrightarrow_\beta^* z$. Indeed: $(\lambda x.x)(\lambda y.y)z \longrightarrow_\beta (\lambda y.y)z \longrightarrow_\beta z$.
- $(\lambda x.y)z \longrightarrow_\beta^* z$. The function ignores its argument: $(\lambda x.y)z \longrightarrow_\beta y$, which is already a normal form.
- ⊗ For every lambda term A , we have $A \longrightarrow_\beta^* A$. The star notation allows zero reduction steps, so every term reduces to itself.
- ‘Lambda’ stands for “Logical Apparatus for Modeling Binding and Dynamic Application”. This is an invented expansion, not the meaning of the notation λ .
- Every lambda term has a normal form. For example, $\Omega = (\lambda x.xx)(\lambda x.xx)$ reduces only to itself and has no normal form.
- ⊗ If a term has a normal form, then it’s unique, up to renaming of bound variables. By the Church–Rosser theorem, two normal forms of the same term must agree up to renaming bound variables.
- ⊗ Let $\bar{3}$ be the Church numeral for 3. Then $\bar{3} F X \longrightarrow_\beta^* F(F(FX))$. By definition, $\bar{3} = \lambda f.\lambda x.f(f(fx))$: it applies its first argument three times to its second.
- ⊗ Let $F = \lambda x.xx$ and $G = \lambda x.\lambda y.yx$. Then $GzF \longrightarrow_\beta^* zz$. Indeed: $GzF \longrightarrow_\beta (\lambda y.yz)F \longrightarrow_\beta Fz \longrightarrow_\beta zz$.

Bonus 1. Since $\text{TRUE} = \lambda x.\lambda y.x$, we have $\text{TRUE FALSE} \longrightarrow_\beta \lambda y.\text{FALSE}$. This is a constant function: it ignores its argument and returns the Boolean FALSE.

Bonus 2. For example, let $A = \text{TRUE}$ and $B = C = \text{ID}$. The two groupings give

$$(AB)C = \text{TRUE ID ID} \longrightarrow_\beta^* \text{ID} = \lambda x.x,$$

whereas

$$A(BC) = \text{TRUE (ID ID)} \longrightarrow_\beta^* \text{TRUE ID} \longrightarrow_\beta^* \lambda y.\lambda x.x = \text{FALSE}.$$

These are not alpha-equivalent.